

Addendum: Exploratory Searches and Numerical Classification

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August 2026

1 Scope

This addendum records numerical findings that are absent or only briefly described in the main write-up. It focuses on brute-force parameter searches, history tests, and numerical continuation. Simulation-based classifications are reported as observations rather than analytic bifurcation results.

2 Constant-Mortality Model

2.1 Initial search for multiple cycles

A history scan at $(K, w) = (1.25, 1.46)$ initially separated the simulations into two candidate cycle clusters. Direct time-series comparisons showed that the candidate behaviors had similar late-time periods and amplitudes. The initial clustering therefore did not provide evidence for two distinct stable limit cycles.

This test identified an important limitation of amplitude-based clustering: slow transients may be classified as different attractors when the integration window is too short. A multiple-cycle claim requires convergence of the full periodic profiles, not only different finite-time maxima and minima.

2.2 Continuation at $w = 1.46$

DDE-BIFTOOL continuation of the constant-mortality model detected a Hopf point at

$$K_H = 0.5931854588. \tag{1}$$

The continued periodic branch developed a rapidly increasing chronological period. The largest successfully computed value was

$$K = 1.2559216155, \tag{2}$$

$$T_{\text{chron}} = 966.817642 \text{ days}, \tag{3}$$

$$T_\phi = 56.774001, \tag{4}$$

$$\frac{dt}{d\phi} = 20.435078, \tag{5}$$

$$x_{\text{min}} = 2.8843690220 \times 10^{-2} \text{ mg C L}^{-1}. \tag{6}$$

The large chronological period results from both an increasing period in developmental time and a slowing of the chronological clock. Continuation near this endpoint produced nearly singular correction matrices, so the numerical branch should not be interpreted beyond the last corrected points. These calculations support a long-period approach to the end of the periodic regime, but do not identify the endpoint bifurcation by themselves.

3 Variable-Mortality Parameter Search

The most successful exploratory parameter set was

$$w = 3, \quad \mu_j = 7.45 \times 10^{-3}, \quad \mu_a = 5.92 \times 10^{-6}. \quad (7)$$

The value of μ_j was selected as an exploratory calibration. Using the Ascoli juvenile death rate and a representative high-food state gives

$$\mu_j \approx \delta_j e_j h_j(Q_{\text{ref}}) f_j(x_{\text{ref}}) \quad (8)$$

$$\approx (0.06)(0.5)(1)(0.25) = 7.5 \times 10^{-3}. \quad (9)$$

Thus, $\mu_j = 7.45 \times 10^{-3}$ is consistent with this reference-state calculation, but it is not an independently measured mortality parameter.

4 Two-History Scan in K

Resource-rich and grazer-rich histories were tested over $0.5 \leq K \leq 6.5$. Each trajectory was integrated to 20,000 chronological days, and its final 5,000 days were classified. The broad result was

Table 1: Simulation classifications for $w = 3$ and the selected mortality scalars.

K range	Observed classification
$0.5 \leq K \leq 1.2$	Both histories were classified primarily as equilibria.
$1.3 \leq K \leq 2.5$	Both histories generally produced the same classified periodic behavior.
$K = 2.6$	Both histories were periodic, but their finite-time resource amplitudes differed.
$K \geq 2.7$	Most simulations were unresolved rather than clearly periodic or clearly equilibrial.

In this classification, “unresolved” means that the final-window amplitude was too large for the equilibrium tolerance but that too few consistent peaks were detected for a cycle classification. It does not identify the asymptotic state. The unresolved simulations may be slowly converging equilibria, long transients, or very long-period oscillations.

Clearly classified cycles disappeared above $K = 2.6$ in this scan. This is consistent with high- K cycle collapse, but the unresolved classifications do not prove convergence to an interior equilibrium.

5 History Dependence Near the Apparent Transition

At $K = 2.6$, the two-history scan produced the measurements in Table 2.

Table 2: Late-time measurements at $K = 2.6$ and $w = 3$.

Quantity	Resource-rich	Grazer-rich
Resource amplitude (mg C L^{-1})	1.2840	0.2052
Grazer amplitude (mg C L^{-1})	0.4351	0.3939
Cycle period (days)	25.44	25.66
Median stage duration (days)	55.64	49.41
$\text{median}(\tau)/T_{\text{cycle}}$	2.187	1.926

The resource amplitudes were substantially different, while the grazer amplitudes differed by approximately ten percent. Both simulations satisfied $\text{median}(\tau) > T_{\text{cycle}}$.

A separate four-history experiment at $K = 2.5$ also produced two finite-time families after 20,000 days. Resource-rich and juvenile-heavy histories gave resource amplitudes near 1.43, whereas grazer-rich and adult-heavy histories gave amplitudes near 0.20. All four periods were approximately 25.53 days. However, the broader two-history K scan produced nearly identical smaller-resource cycles from both histories at $K = 2.5$. The two experiments therefore did not give a reproducible multiple-attractor result.

The current evidence is insufficient to distinguish coexisting stable cycles from history-dependent long transients near the apparent transition. A stable multiple-cycle result would require two distinct periodic profiles that retain fixed amplitudes and periods under longer integration and perturbation.

6 Stage Duration Relative to Period

For clearly classified variable-mortality cycles over approximately $1.3 \leq K \leq 2.6$, representative values were

$$T_{\text{cycle}} \approx 23\text{--}26 \text{ days}, \quad (10)$$

$$\text{median}(\tau) \approx 49\text{--}56 \text{ days}, \quad (11)$$

$$\frac{\text{median}(\tau)}{T_{\text{cycle}}} \approx 1.9\text{--}2.2. \quad (12)$$

This establishes that the variable-mortality model can produce cycles for which stage duration exceeds population period. It does not establish the coexistence of the large- and small-amplitude branches reported in the Nature model.

7 History Extraction and Long Transients

Constant histories produced long approach times in several simulations. To display an established cycle from $t = 0$, a complete history segment of length one in developmental time was extracted from a later periodic portion of a first solve and used to initialize a second solve. For the chronological clock T and cumulative survival state R , the restarted histories were shifted as

$$\tilde{T}(\phi) = T(\phi_* + \phi) - T(\phi_*), \quad (13)$$

$$\tilde{R}(\phi) = R(\phi_* + \phi) - R(\phi_*), \quad -1 \leq \phi \leq 0. \quad (14)$$

These shifts preserve the delayed differences that determine stage duration and survival. The restarted simulation displayed the selected periodic behavior immediately. This method is appropriate for plotting or following a known cycle; because it intentionally initializes near that cycle, it does not measure the size of its basin of attraction.

8 Numerical Qualifications

The transformed solver runs in developmental time ϕ , not chronological days. Since

$$\frac{dt}{d\phi} = \frac{1}{\rho D_j(\phi)}, \quad (15)$$

a large endpoint in ϕ can represent a much longer chronological simulation. Chronological event functions are therefore preferable when runs must be compared at equal durations.

For the variable-mortality model, survival was evaluated using a cumulative state $R'(\phi) = D_j(\phi)^{-2}$ and the difference $R(\phi) - R(\phi - 1)$. At very long times, subtracting two large cumulative values may lose numerical precision. This should be checked when interpreting small late-time amplitudes.

9 Conclusions Supported by the Searches

The exploratory calculations support three conclusions:

1. The constant-mortality model exhibited a long-period endpoint of its continued periodic branch, but the initial history clustering did not verify multiple stable cycles.
2. The variable-mortality parameter set $(w, \mu_j, \mu_a) = (3, 7.45 \times 10^{-3}, 5.92 \times 10^{-6})$ produced cycles with stage duration greater than cycle period and a loss of clearly classified cycles at higher K .
3. History-dependent finite-time amplitudes occurred near the apparent transition, but coexisting stable limit cycles were not established.

Direct confirmation of the Nature-type structure would require continuation of periodic solutions and their stability, including any unstable periodic branch separating two stable cycles.